Indian Pre-Owned Car Price Prediction

Capstone Project Summary | Machine Learning Pipeline (Component 1A) and Streamlit App (Component 1B)

1. Objective

Build an end-to-end machine learning pipeline that predicts the price of a used car in India from its characteristics (maker, model, location, distance driven, age, engine details, fuel type, transmission and more), and deploy it as an interactive Streamlit web application.

2. Work done

Data	Loaded the training set (with Price), the test set (without Price) and the sample submission. Training data has about 42,800 cars and the test set has 5,615 cars.
EDA	Checked missing values, duplicates, data types, price distribution, outliers (IQR rule and boxplots), distributions of distance, age and engine power, category counts, correlations with price, and how fuel type, owner type, transmission, age and distance affect price.
Data cleaning	Removed 117 rows with impossible prices (below Rs. 50,000). Dropped ID (identifier), body_type (mostly missing) and manufacture_year (duplicate of Age of car).
Preprocessing	Split the data 80/20 into training and validation before fitting. Used one reusable ColumnTransformer: median imputation and standard scaling for numeric columns, most-frequent imputation and one-hot encoding for categorical columns (Maker, model, Location, Owner Type, transmission, fuel_type).
Models	Baseline (mean price), Linear Regression, Decision Tree, Random Forest, XGBoost and LightGBM.
Tuning	5-fold cross-validation, and RandomizedSearchCV for Random Forest and XGBoost using cross-validated MAE.
Evaluation	Compared all models on the same validation set using RMSE, MAE and R-squared.
Final model	Tuned XGBoost trained on the log of the price (prices are heavily skewed) and converted back to rupees for predictions. Tree-based models clearly beat Linear Regression and the baseline.
Predictions	Test predictions were generated with the same fitted preprocessor and saved in the required ID, Price format (my_submission.csv).

3. Streamlit app (Component 1B)

- Sidebar inputs: maker, model, location, fuel type, transmission, owner type, distance and age sliders, engine details, seats, doors and audit rating.
- The model and preprocessor are loaded with joblib and st.cache_resource for instant predictions.
- Valuation banding: shows the estimated price with a fair price range of plus or minus 5%.
- Batch Processing tab: upload the test CSV, get bulk predictions and download them as a CSV.
- Insights tab: native Streamlit charts showing how age, distance, fuel type and maker affect price.